CURRICULUM VITAE

Alberto Boffi

Computer science engineer specializing in crafting scalable full-stack applications and and developing AI-driven solutions. Combining a deep mathematical background with an architectural foundation spanning data science, computer graphics, cybersecurity, advanced algorithms and several computing domains.

Personal Data

Name: Alberto Boffi
Date of Birth: 15 August 1999
Current City: Milan, Italy
Citizenship: Italian

Contacts

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WORK EXPERIENCE

May 2024 - Present — Software Engineer, *Connect Reply* (Milan, Italy) - Full time.

- Joined as a hands-on software engineer on a large-scale multi-national web ecosystem of general-purpose connected devices.
- Became a key point of reference for the application's front-end, now deployed across 7+ countries to serve ∼500K users.
- Owner and developer of an upcoming application for the USA market, targeting 1M users.
- Design and implement complex features for data exchange across physical devices, UIs, and BE services, leveraging autonomous agents and LLM-driven solutions.
- Lead the full SDLC and interface with global stakeholders to deliver a robust and efficient E2E delivery process.
- Cultivated strong data science competencies to drive data-informed decision-making.

June 2017 - Sept. 2017 — Software Engineer, *Il Biancospino SCS* (Como, Italy) - Internship.

- Chief the E2E development of a web management software, leveraging a full-stack architecture based on Angular, PHP, and SQL.
- Covered the full SDLC, from requirements analysis and implementation to quality assurance testing.

June 2016 - July 2016	Software Developer, *Popcorn Creative Lab* (Como, Italy) - Internship. • Developed institutional websites and web applications for tourism and business sectors. • Cultivated valuable skills in the field of graphic design.

EDUCATION

Sept. 2021 - April 2024	Politecnico di Milano, *Master of Science*, Computer Science and Engineering. **Graduation grade**: 110/110 cum laude. **Thesis title**: VirtualRelativity: An Interactive Simulation of the Special Theory of Relativity in Virtual Reality.
Sept. 2018 - Sept. 2021	Politecnico di Milano, *Bachelor of Science*, Ingegneria Informatica.
Sept. 2013 - June 2018	I.T.I.S. Magistri Cumacini, *High School Diploma*, Informatica e Telecomunicazioni. **Thesis project**: Fluo, a web-based solution that enables users to generate professional-grade websites via an intuitive graphical interface and dynamic UI-driven configuration.

SKILLS

FRONT-END DEVELOPMENT

Web development and experience in diverse web technologies.

Research focus on Progressive Web Applications.

Technical deep dive into the Angular ecosystem and its architectural patterns.

Active development of scalable applications using ReactJs and Redux.

Ongoing exploration and prototyping with Svelte for lightweight web solutions.

High knowledge of UI/UX principles, encompassing User-Centered Design, visual hierarchy, and the creation of accessible, responsive interfaces.

Proficiency in Three.js for 3D web rendering and interactive environments.

Strong proficiency in TypeScript, modern JavaScript (ES6+), and semantic HTML5.

Advanced CSS3 and Sass for scalable, maintainable, and responsive styling.

BACK-END DEVELOPMENT

Proficiency in Node.js and PHP for server-side applications.

Experience with Java, Python and C# across different software projects.

Design and implementation of RESTful APIs, GraphQL services and gRPC-based communication.

Experience with event-driven architectures and event streaming systems.

Implementation of Identity and Access Management (IAM), authentication and web security mechanisms.

Design and management of relational databases using SQL/MySQL, including schema design and query optimization.

Knowledge of NoSQL database models, including document, key-value and graph databases.

Experience with microservices architectures and distributed systems.

Scripting and automation using Lua.

DEVOPS

Full stack development of high-impact MVPs, balancing rapid prototyping with architectural scalability.

Application of software design patterns and delivery of high-quality, well-documented code.

Design of scalable software architectures, deployment environments and Docker-based containerization.

Setup and configuration of error-tracking and analytics platforms, including Sentry, New Relic, Google Analytics and Google Search Console

Management of the Software Development Lifecycle (SDLC), from planning to deployment.

Project management using Agile methodologies (Scrum/Kanban).

Planning and implementation of testing strategies, including unit, functional, E2E, smoke and UAT testing.

Version control and repository management using Git and GitHub.

CI/CD pipelines, source code automation and release management using Azure DevOps.

Project and technical documentation management with Jira and Confluence.

Experience with event-driven architectures.

Strong focus on software performance, scalability, usability and non-functional requirements (NFRs).

ARTIFICIAL INTELLIGENCE

Development of Machine Learning solutions using Scikit-learn for predictive modeling and data analysis.

Design and training of Deep Learning models with TensorFlow for complex learning tasks.

Integration of Large Language Models into modern software architectures and intelligent applications.

Development of AI-powered applications through Hugging Face and LangChain frameworks.

Implementation of Retrieval-Augmented Generation (RAG) architectures using vector databases for semantic retrieval.

Design of agentic AI workflows based on the Model Context Protocol (MCP).

Application of prompt engineering techniques for reliable and context-aware LLM interactions.

Research and experimentation with Reinforcement Learning (Gymnasium).

DATA SCIENCE

Design of relational data models, database schemas and data normalization strategies.

Development of ETL pipelines for reliable data integration and transformation processes.

Exploratory Data Analysis (EDA) and analytical techniques for data-driven decision making.

Design and evaluation of A/B testing methodologies for experimental validation.

Application of data security principles and best practices for secure data management.

AUTOMATION

Streamlining of workflows using Claude Code, GitHub Copilot, and specialized CLI tools.

Advanced development with AI-native environments like Cursor, Google Antigravity and Google Jules.

Rapid prototyping and MVP development via Google AI Studio, Lovable and Google Opal.

EMBEDDED SYSTEMS

Digital hardware design using VHDL for FPGA-based architectures and digital circuit implementation.

Development of Internet of Things (IoT) applications for connected and distributed systems.

Embedded software development using C and C++ for low-level hardware interaction.

Design and programming of Single-Board Computer (SBC) systems for embedded applications.

Knowledge of computer hardware architectures, processor organization and low-level system design.

Fundamental understanding of electrical engineering principles.

Solid foundation in computer networking principles and communication protocols.

CYBERSECURITY

Cryptography and secure communication protocols.

Authorization and access control systems.

Proficiency in the fundamental principles of secure software design.

Main focus on web security and risks related to web applications.

MEDIA

UI/UX design and interactive prototyping using Figma and Canva for modern digital interfaces.

Creation and editing of raster and vector graphics, ranging from technical illustrations to digital assets.

Production and post-processing of multimedia content, encompassing video editing and visual storytelling.

Development of Virtual Reality experiences using the Unity engine.

Integration of AI-assisted visual content generation using Google Stitch and Google Pomelli.

MATHEMATICS

Advanced study of algorithms, data structures and theoretical computer science.

Study of computer graphics, rendering pipelines and geometric transformations for interactive applications.

Application of operations research and mathematical optimization techniques for complex decision-making problems.

Solid background in discrete mathematics, formal logic and algebra, supporting rigorous computational reasoning.

Deep theoretical understanding of calculus and physics for scientific applications.

SOFT SKILLS

Strong background in market analysis and go-to-market strategies.

Expertise in stakeholder management, ensuring technical alignment with business goals through direct client relations and frequent international off-site missions.

Ability to bridge the gap between complex engineering constraints and strategic business requirements.

Good knowledge of additional project-management-platforms (Notion, Monday or Trello, etc.) and productivity ecosystems (Microsoft Office Suite, Google Workspace).

MASTER'S COURSES

30/30 cum laude	Foundations of Operations Research
30/30	Logica e Algebra 2 (Advanced topics in Algebra and Mathematical Logic)
30/30	Software Engineering 2 (Advanced topics in Software Engineering)
30/30	Data Bases 2 (Advanced topics in Data Bases)
27/30	Foundations of Artificial Intelligence
30/30	Philosophical Issues of Computer Science
30/30	Technologies for Information Systems
27/30	Computer Security
28/30	Digital Forensics and Cybercrimes
30/30 cum laude	Game Theory
30/30	Computing Infrastructures
30/30	Computer Ethics
28/30	Formal Languages and Compilers
26/30	Artificial Neural Networks and Deep Learning

27/30	Business for Information Systems
30/30 cum laude	Computer Graphics
26/30	Advanced Computer Architectures
30/30 cum laude	Discrete Mathematics
21/30	Numerical Analysis
30/30	Advanced Algorithms and Parallel Programming

TECHNICAL WRITING

- *Progressive Web Apps: The Web's Answer to Native Applications.*
  DEV Community Series [PDF version]
  Applied research to evaluate KPIs of PWAs for potential adoption in enterprise projects.

- *From One-Time-Pad to PKI: How Real-World Encryption Protects Our Data.*
  DEV Community Series [PDF version].
  Highly technical review of cryptographic techniques and their theoretical foundations.

- *Move Fast, Fail Fast, Judge Fair: Why Tech Needs Outcome-Based Responsibility.*
  DEV Community Article [PDF version].
  Comparative analysis of responsibility attribution frameworks and their impact on modern engineering teams.

- *Boffi, A., Puppin, E., & Contran, M. (2024). VirtualRelativity: An Interactive Simulation of the Special Theory of Relativity in Virtual Reality.*
  arXiv Preprint [Full thesis].
  Master's Thesis in the field of Virtual Reality applied to Einstein's Special Theory of Relativity.

- *Boffi, A. (2021). An efficient way to manage ranges of data with Wise Red-Black Trees.*
  arXiv Preprint.
  Investigation and benchmarking of a custom data structure designed to optimize academic project performance.

PERSONAL PORTFOLIO HIGHLIGHTS

My academic and personal efforts mainly focus on software engineering, applied mathematics and applied physics. Below is a non-exhaustive selection of these projects.

Some previously launched projects are no longer maintained or publicly available, while others, such as VirtualRelativity and various corporate solutions, remain active and in use today. To replace archived work, I am actively building new MVPs currently in BETA or early development. This list excludes most of my enterprise projects: they eat up most of my time, but they cannot be disclosed publicly due to NDA restrictions.

To check out my latest public repositories and active work, please check out my Github profile.

RELEASED PROJECTS

April 2024

VirtualRelativity: An Interactive Simulation of the Special Theory of Relativity in Virtual Reality • [*Live*]

Github ∘ Website ∘ Demo ∘ Use Case ∘ Learn More

VirtualRelativity is an innovative VR solution designed to simulate the effects of Einstein's Special Theory of Relativity in a digital environment. The current implementation includes a Unity package that accurately models space contraction, time dilation, and the relativistic Doppler effect, as well as a final application that provides engaging ways to understand relativistic phenomena through real-life scenarios.

The application has also been used by Master's students in Physical Engineering at Politecnico di Milano, and the work has started drawing citations across multiple physiscs sources, paving the way for future development in scientific education.

March 2022

Automatic Configurator of Simulation Environments for Human-Robot Interaction in Healthcare Scenarios • [*Archived*]

Technical Report

Simulation framework developed in collaboration with researchers and professors at Politecnico di Milano to model and evaluate stochastic human-robot interaction within healthcare environments. Designed to bridge the gap between robotics engineering and clinical workflows, the system enabled healthcare professionals to define and customize interaction scenarios through high-level declarative notation without low-level programming. The framework automatically generated corresponding 3D virtual environments, delivering real-time visual simulation and quantitative evaluation to assess patient-robot safety, workflow efficiency, and collaborative dynamics.

Feb. 2019

Friscool: Note Sharing Platform • [*Archived*]

Demo

Friscool was a web application that I developed during my first year of university, in order to enable students to share and access lecture notes across courses. The aim was to promote an exclusive way to collaborate and exchange knowledge within the academic community.

June 2018

Fluo: Website Composer • [*Archived*]

Demo

Fluo was a drag-and-drop tool for website creation that I developed as a high-school thesis project. It was designed as an All-in-One Workspace targeting non-technical users, whose main features included multi-site hosting, cloud storage, a file system for website and page management, object manipulation through snapping guides, real-time auto-saving, comprehensive text formatting, and position and scale control of images, text, and geometric shapes, both through numerical input and manual manipulation.

UPCOMING MVPs

- Moka PHP: A Lightweight PHP Library for High-Performance REST APIs • [*In Progress*]

  Github ∘ Website

  As my newest project, currently in its early stages of development, Moka PHP is built to overcome the overhead of monolithic frameworks as a lightweight library engineered for ultra-low latency REST APIs and developer simplicity. With an intuitive, clutter-free setup, it is ideal for high-throughput applications, AI microservices requiring real-time agent/LLM data streaming, and rapid prototyping where speed and ease of use are non-negotiable. Development is currently focused on expanding the core to include a dedicated abstraction layer for SQL database interactions and a comprehensive module for user authentication and account management.

- Ludico: Web Platform for Instant Party & Board Games • [*BETA*]

  Live Usage

  Designed to eliminate the friction of traditional party games, Ludico is a real-time web platform engineered for instant, multiplayer entertainment. Built around a modular game engine and an event-driven architecture, it leverages optimized client-server messaging and state synchronization to deliver seamless, low-latency social play across devices without requiring app downloads or account setup. It currently features "Mind Link", already a top-tier digital adaptation of the popular Italian show Reazione a Catena ("Intesa Vincente"), and the newly released beta "The Impostor". Current development is mainly focused on infrastructure scaling.

- Next-Gen Digital Pedagogy via Interactive STEM Technologies • [*MVP*]

  GitHub ∘ Demo ∘ Live Usage

  My vision is to modernize and digitize STEM education by replacing static, theoretical instruction with interactive learning platforms. To demonstrate the viability of this vision, the first project in this initiative is a realistic 3D Solar System simulator. Designed to balance astronomical accuracy with educational clarity, the application features tailored planetary scaling, orbital mechanics, dynamics, and realistic lighting. By allowing users to freely explore celestial bodies in real time, the application demonstrates how interactive technology can simplify complex natural systems and reshape science learning.

ADDITIONAL RESEARCH WORKS

- Research Collaboration, 2022, *Autocorrelation Analysis For Fluorescence Correlation Spectroscopy*

  Collaboration with university faculty on academic research aimed at optimizing Fluorescence Correlation Spectroscopy (FCS) implementations through GPU acceleration. Focused on analyzing and enhancing multi-tau algorithms for autocorrelation computation, while conducting a comprehensive review of state-of-the-art techniques and performance benchmarks in similar GPU-accelerated FCS systems.

TRAINING COURSES

Stanford University	Special Relativity & Classical Field Theory.
Stanford - SOE-YCS0009	Divide and Conquer, Sorting and Searching, and Randomized Algorithms.

LANGUAGES

Italian	Native.
English	Proficient. Latest certificate: B2 Level, *TOEIC L&R*, 07/29/2021.

EXTRACURRICULAR ACTIVITIES

Sep. 2021 - June 2022	Member of the Official Athletics Team of Politecnico di Milano.
Aug. 2006 - Feb. 2021 Aug. 2026 - Present	Competitive football career spanning 15 years, with experience up to the regional level.

In compliance with the Italian Legislative Decree no. 196 dated 30/06/2003, I hereby authorize the recipient of this document to use and process my personal details for the purpose of recruiting and selecting staff and I confirm to be informed of my rights in accordance to art. 7 of the above mentioned decree.